



POLICY

Medication: Lithium – Prescribing and Management

Scope (Staff):	North Metropolitan Health Service Mental Health (NMHS MH) Clinical Staff
Scope (Area):	NMHS MH All Areas

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1. Aim

This document provides NMHS MH clinical staff with the requirements for prescribing and management of lithium to ensure safe and effective treatment for patients of the service.

2. Background

Lithium is a High Risk Medication as defined by the [WA Health High Risk Medication Policy](#).

High risk medications will be managed by NMHS MH services according to evidence- based practice in order to ensure patient safety and therapeutic outcomes ([NMHS MH High Risk Medication Policy](#)).

3. Risk

Patients may suffer adverse outcomes related to lithium if it is not prescribed and managed in accordance with this policy.

Governance & Accountability	☒	Policy & Legislative Compliance	☒
	☐	Quality of Patient Care	☒

4. Principles

4.1 Formulations

- 250mg lithium carbonate immediate release tablet (Lithicarb®)
- 450mg lithium carbonate slow release tablet (Quilonum®)
- 127mg lithium citrate BP per 1mL oral solution. (Auspman®)

127mg Lithium citrate = 50mg Lithium carbonate.

Lithium citrate oral solution is equivalent to 50mg/mL lithium carbonate

4.2 Indications

Lithium is indicated for the treatment of:

- acute mania
- acute bipolar depression
- prophylaxis of bipolar disorder
- as augmentation therapy in depression, schizoaffective disorders and chronic schizophrenia.

Lithium has been used off-label to increase the white cell count of patients receiving clozapine therapy. Lithium used for this and other off-label indications should only be prescribed by a consultant psychiatrist.

4.3 Commencement

Before commencing lithium therapy baseline assessments must include:

- Renal function
- Full blood picture
- Thyroid function (including TSH)
- Serum electrolytes
- Serum calcium (including ionised calcium)
- Electrocardiogram (ECG) for patients aged below 16 or above 65 years old, patients with cardiac disease, on other medications which may prolong QT interval and when clinically indicated
- Pregnancy test as appropriate. Lithium does carry a small risk of congenital heart malformation (Ebstein's anomaly). Reliable contraception should be used by women of childbearing age. If a patient becomes pregnant on lithium, a risk/benefit assessment should be performed with regards to continuing treatment.

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- Baseline weight (desirable)

All patients commenced on lithium at all NMHS MH Sites must be provided with written information on lithium treatment and the provision of this education recorded in the patient's integrated progress notes (PMR 7).

4.4 Dosing and Therapeutic Drug Monitoring

Lithium dose is titrated according to patient response and plasma levels. Consider all interacting medication when prescribing lithium and if necessary adjust dose and/or monitoring.

The recommended therapeutic serum lithium range for:

- ACUTE mania is between 0.5 -1.2 mmol/L.
- MAINTENANCE treatment is between 0.5 -1.0 mmol/L.
- In the elderly (>65 years) is between 0.4 - 0.7mmol/L.

In addition to age-related decline in renal function increasing the risk of lithium toxicity, older people may be more sensitive to its toxic effects; use lower doses and monitor more frequently.

Serum levels should be taken 5-7 days after commencement of lithium or after a dose change.

Venepuncture to determine serum lithium levels should be performed 12 hours after the last dose. Doses that would have been given immediately prior to blood being drawn should be withheld until after venepuncture.

4.5 Withdrawal/Cessation of lithium

Abrupt cessation of lithium can lead to a relapse of mania (or, less likely, depression) within a few months. Therefore, it should be withdrawn slowly over at least 2-3 months, if possible.

4.6 Physical Monitoring

On admission to hospital all patients on lithium therapy must receive the following laboratory tests, unless results from tests in the previous 7 days are available:

- Renal function
- Full blood picture
- Thyroid function (including TSH)
- Serum electrolytes
- Serum lithium level
- Serum calcium (including ionised calcium). Parathyroid hormone concentrations should be measured if calcium concentrations are raised.

The following tests should be performed during maintenance treatment and when clinically indicated:

- Serum lithium levels every 6 months after a stable therapeutic concentration is achieved



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- Renal function & serum electrolytes every 6 months
 - Thyroid function (including TSH) every 12 months
 - To screen for hyperparathyroidism, calcium concentrations should be measured at least annually. (Parathyroid hormone concentrations should be measured if calcium concentrations are raised).
 - Body weight/Body Mass Index at least annually.

4.7 Lithium Toxicity

Lithium is a narrow therapeutic index drug. It is handled in the body in a similar way to sodium; many risk factors for toxicity relate to changes in sodium and fluid status.

The most important factors predisposing to lithium toxicity are:

- Interactions with medications e.g. drugs that can affect renal function (eg diuretics, non-steroidal anti-inflammatory drugs (NSAIDs), Angiotensin-Converting Enzyme (ACE) Inhibitors and Angiotensin II receptor blockers). Refer to clinical pharmacists, the Graylands Drug Information service or references for consideration of potentially interacting drugs.
- Dehydration
 - Reduced fluid intake
 - Fluid loss from vomiting, diarrhoea or excessive sweating
- Renal dysfunction
- Low sodium levels
- High starting dose, deliberate or inadvertent overdose

Symptoms of toxicity are common when the lithium concentration is above 1.5 mmol/L, however they can occur at lower concentrations, particularly in the elderly. Urgent lithium levels and Urea & Electrolytes are to be ordered if lithium toxicity is suspected.

Patients should be monitored for, and advised of, the signs and symptoms of lithium toxicity.

4.7.1 Mild to moderate toxicity

The early signs and symptoms are either:

- Gastrointestinal: increasing anorexia, nausea and diarrhoea; or
- Central Nervous System effects: muscle weakness, drowsiness, ataxia, coarse tremor and muscle twitching, irritability and reduced cognition.

4.7.2 Severe Toxicity

When lithium blood levels exceed 2 mmol/L symptoms commonly include:

- Increased disorientation
- Increased muscle tone
- Myoclonic jerks and seizures, with

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- Progression to coma and ultimately death.

5. Compliance and Evaluation

Compliance with this policy will be audited annually by NMHS MH Pharmacy services:

- Zero reports on Datix CIMS involving non-adherence to this Lithium policy.
- 100% of patients treated with lithium will have documented evidence of receiving written information on commencement.
- 100% of patients treated with lithium will be monitored in accordance with the policy.

Related internal policies, procedures and guidelines
NMHS MH Medication: Management of High Risk Medication Policy

References
WA Health High Risk Medication Policy MP 0131/20 Australian Medicines Handbook (AMH) Maudsley Prescribing Guidelines in Psychiatry 14 th Edition e-Therapeutics Guidelines: Psychotropic Mental Health Act 2014

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NSQHS Standards Applicable:	☒  Std 1: Clinical Governance		☒  Std 4: Medication Safety		
National Standards for Mental Health Services	☐ Std 2: Safety ☒ Std 3: Consumer and Carer Participation ☒ Std 5: Promotion and Prevention		☒ Std 8: Governance, leadership and management ☒ Std 10: Delivery of Care		
Chief Psychiatrist's Standards for Clinical Care:	Physical Health Care of Mental Health Consumers, Risk Assessment and Management, Transfer of Care				
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8/09/2017	1.0	NMHS MHEG	Out of session endorsement via email
15/09/2017	1.0	NMHS MHPH Executive director	Submission form
15/09/2017	1.0	NMHS MH Chief Pharmacist	Initial endorsement; replaces Lithium: Prescribing Guidelines
18/05/2018	1.1	A/Policy Officer	HealthPoint hyperlinks repaired. Review date remains 15/09/2020
17/07/2020	2.0	NMHS MH Chief Pharmacist	Template updated; risk statement added.
26/06/2023	3.0	NMHS MH Chief Pharmacist	Template update; information reviewed. Formulation information added.

The health impact upon Aboriginal people have been considered, and where relevant incorporated and appropriately addressed in the development of this health initiative (ISD Number: 2315).

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